

NEET 2026 SAMPLE PAPER 02

1. Concentrated nitric acid is 70% HNO_3 . The amount of concentrated nitric acid solution that should be used to prepare 250 mL of 2.0 M HNO_3 would be:

- A. 90.0 g conc. HNO_3
- B. 70.0 g conc. HNO_3
- C. 54.0 g conc. HNO_3
- D. 45.0 g conc. HNO_3

2. 6.02×10^{20} molecules of urea are present in 100 mL of its solution. The concentration of the solution is:

- A. 0.02M
- B. 0.01M
- C. 0.001M
- D. 0.1M

3. The de-Broglie wavelength of a particle having the momentum of $3.3 \times 10^{-24} \text{ kg m s}^{-1}$ will be

- A. 0.002 Å
- B. 0.02 Å
- C. 0.2 Å
- D. 2 Å

4. Which of the following statements is not true with respect to photoelectric effect

- A. The plot of kinetic energy of the photoelectrons against frequency of absorbed radiation gives a straight line parallel to x-axis
- B. The plot of K.E. of the photoelectrons against frequency of absorbed radiation gives a straight line with slope equal to h and intercept to work function of metal
- C. The plot of kinetic energy of the photoelectrons versus the intensity of incident radiation with constant frequency gives a straight line parallel to x-axis
- D. When intensity of light is increased the number of photons ejected increases but the energies of these electrons are the same.

5. Which of the following order is not in accordance with the property stated against it?

- A. $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$:Oxidising Power
- B. $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$:Acidic property in water
- C. $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$:Electronegativity
- D. $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$:Bond dissociation energy

6. An element R forms the highest oxide R_2O_5 . R belongs to:

- A. 13th group
- B. 15th group
- C. 16th group
- D. 2nd group

7. Which of the following would have a permanent dipole moment?

- A. BF_3
- B. SiF_4
- C. SF_4
- D. XeF_4

8. In a regular octahedral molecule MX_6 , the number of X-M-X bonds at 180° is

- A. 3
- B. 2
- C. 6
- D. 4

9. One mole of a non-ideal gas undergoes a change of state

$(2.0 \text{ atm}, 3.0 \text{ L}, 95 \text{ K}) \rightarrow (4.0 \text{ atm}, 5.0 \text{ L}, 245 \text{ K})$ with a change in internal energy, $\Delta U = 30 \text{ L atm}$. The change in enthalpy (ΔH) of the process in L atm is:

- A. 40.0
- B. 42.0
- C. 44.0
- D. not defined, because pressure is not constant

10. ΔS will be highest for the reaction:

- A. $\text{Ca(s)} + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{CaO(s)}$
- B. $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO(s)} + \text{CO}_2(\text{g})$
- C. $\text{C(s)} + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$
- D. $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO(g)}$

11. What is the relationship between the degree of dissociation (α) of PCl_5 and the pressure at equilibrium in the reaction: $\text{PCl}_5(\text{g}) \rightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$?

- A. $\alpha \propto P$
- B. $\alpha \propto \frac{1}{\sqrt{P}}$
- C. $\alpha \propto \frac{1}{P^2}$
- D. $\alpha \propto \frac{1}{P^4}$

12. Which salt undergoes hydrolysis?

- A. CH_3COONa
- B. KNO_3
- C. NaCl
- D. K_2SO_4

13. Redox reaction among the following is

- A. $\text{NaCl} + \text{KNO}_3 \rightarrow \text{NaNO}_3 + \text{KCl}$
- B. $\text{CaC}_2\text{O}_4 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{C}_2\text{O}_4$
- C. $\text{Mg}(\text{OH})_2 + 2\text{NH}_4\text{Cl} \rightarrow \text{MgCl}_2 + 2\text{NH}_4\text{OH}$
- D. $\text{Zn} + 2\text{AgCN} \rightarrow 2\text{Ag} + \text{Zn}(\text{CN})_2$

14. Which of the following compounds is the weakest Lewis acid?

- A. BF_3
- B. BCl_3
- C. BBr_3
- D. BI_3

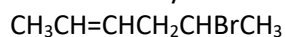
15. Name the type of the structure of silicate in which one oxygen atom of $[\text{SiO}_4]^{4-}$ is shared?

- A. Sheet silicate
- B. Pyrosilicate
- C. three dimensional silicate
- D. Linear chain silicate

16. Which compound contains an isopropyl group?

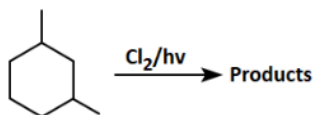
- A. 2,2,3,3-Tetramethylpentane
- B. 2,2-Dimethylpentane
- C. 2,2,3-Trimethylpentane
- D. 2-Methylpentane

17. How many stereoisomers does this molecule have?



- A. 4
- B. 6
- C. 8
- D. 2

18. How many monochlorinated products (excluding stereoisomers) are obtained from the given reaction?



- A. 4
- B. 5
- C. 6
- D. 7

19. Which of the following colligative properties is associated with the concentration term 'molarity'?

- A. Lowering of vapour pressure
- B. Osmotic pressure
- C. Depression in freezing point
- D. Elevation in boiling point

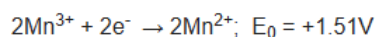
20. Which of the following aqueous solution has osmotic pressure nearest to that an equimolar solution of $K_4[Fe(CN)_6]$?

- A. Na_2SO_4
- B. $BaCl_2$
- C. $Al_2(SO_4)_3$
- D. $C_{12}H_{22}O_{11}$

21. A current is passed through two voltmeters connected in series. The first voltmeter contains $XSO_4(aq)$ while the second voltmeter contains $Y_2SO_4(aq)$. The relative atomic masses of X and Y are in the ratio of 2:1. The ratio of the mass of X liberated to the mass of Y liberated is:

- A. 1:1
- B. 1:2
- C. 2:1
- D. None of the above

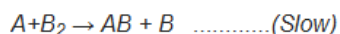
22. Given below are two half-cell reactions:



The E_0 for $3Mn^{2+} \rightarrow 2Mn^{3+} + Mn$ will be:

- A. -2.69V; the reaction will not occur
- B. -2.69V; the reaction will occur
- C. -0.33V; the reaction will not occur
- D. -0.33V; the reaction will occur

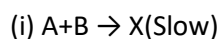
23. A hypothetical reaction, $A_2 + B_2 \rightarrow 2AB$ mechanism as given below;



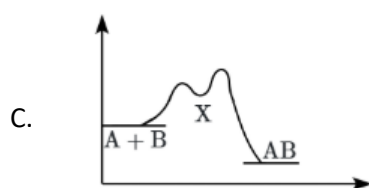
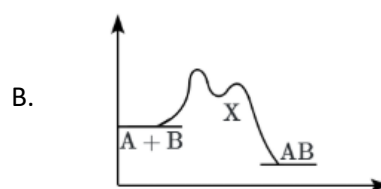
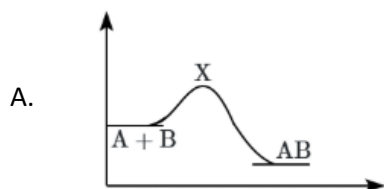
The order of the overall reaction is

- A. 2
- B. 1
- C. 1.5
- D. Zero

24. For an exothermic chemical process occurring in two steps as;



The progress of the reaction can be best described by:



D. All of the above.

25. Among the following reactions, N_2 is not generated by heating:

- A. $(NH_4)_2Cr_2O_7$
- B. $NH_4Cl + NaNO_2$
- C. $NH_4Cl + CaO$
- D. $Ba(N_3)_2$

26. Strong reducing behaviour of H_3PO_2 is due to

- A. Presence of one-OH group and two P-H bonds
- B. High electron gain enthalpy of phosphorus
- C. High oxidation state of phosphorus
- D. Presence of two-OH groups and one P-H bond

27. At $pH = X$, CrO_4^{2-} (yellow) changes to $Cr_2O_7^{2-}$ (orange), and vice versa at $pH = Y$. What are the values of X and Y, respectively?

- A. 6, and 8
- B. 6, and 5
- C. 8, and 6
- D. 7 and 7

28. A diamagnetic lanthanoid ion among the following is:

(At. numbers: Ce = 58, Sm = 62, Eu = 63, Yb = 70)

- A. Ce^{2+}
- B. Eu^{2+}
- C. Sm^{2+}
- D. Yb^{2+}

29. The coordination number and oxidation state of Cr in $K_3[Cr(C_2O_4)_3]$ are respectively

- A. 3 and +3
- B. 3 and 0
- C. 6 and +3
- D. 4 and + 2

30. The octahedral complex will not show geometrical isomerism is:
(A and B are monodentate ligands)

- A. $[MA_4B_2]$
- B. $[MA_5B]$
- C. $[MA_2B_4]$
- D. $[MA_3B_3]$

31. What is produced when Br_2 is added to cis-but-2-ene?

- A. Racemic mixture of 2,3-Dibromobutane
- B. Meso form of 2,3-Dibromobutane
- C. Dextro form of 2,3-Dibromobutane
- D. Laevo form of 2,3-Dibromobutane

32. Iodoform can be prepared from all except

- A. Isopropyl alcohol
- B. 3-methyl-2-butanone
- C. isobutyl alcohol
- D. ethyl methyl ketone

33. Which set of reactants among the following produces anisole?

- A. CH_3CHO , $RMgX$
- B. C_6H_5OH , $NaOH$, CH_3I
- C. C_6H_5OH , neutral $FeCl_3$
- D. $C_6H_5-CH_3$, CH_3COCl , $AlCl_3$

34. The reaction of ethyl formate with an excess of CH_3MgI followed by hydrolysis gives:

- A. n-Propyl alcohol
- B. Ethanol
- C. Isopropyl alcohol
- D. Propanal

35. A carbonyl compound reacts with hydrogen cyanide to form cyanohydrin which on hydrolysis forms a racemic mixture of α -hydroxy acid. The carbonyl compound is

- A. acetaldehyde
- B. acetone
- C. diethyl ketone
- D. formaldehyde

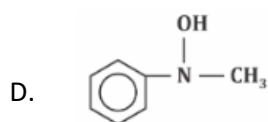
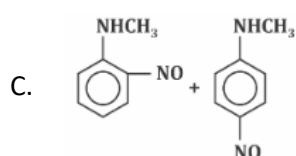
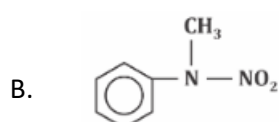
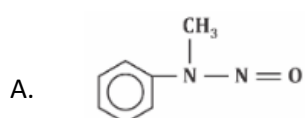
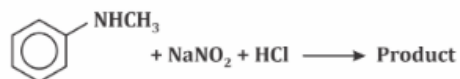
36. The compound which on reduction with LiAlH_4 gives two different alcohols:

- A. $\text{CH}_3\text{COOCH}_3$
- B. $\text{CH}_3\text{COOC}_2\text{H}_5$
- C. CH_3COCH_3
- D. CH_3CHO

37. Acetamide and ethyl amine can be distinguished by reacting with:

- A. Aq. HCl and heat
- B. Aq. NaOH and heat
- C. Acidified KMnO_4
- D. Bromine water

38. Predict the product,



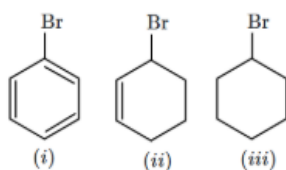
39. α -D-(+)-glucose and β -D-(+)-glucose are

- A. Anomers
- B. Epimers
- C. Enantiomers
- D. Geometrical isomers

40. Which type of amino acid is lysine?

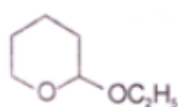
- A. β - amino acid
- B. Acidic
- C. Basic
- D. Neutral

41. The increasing order of hydrolysis of the following compounds:



- A. (i) < (iii) < (ii)
- B. (ii) < (iii) < (i)
- C. (ii) < (i) < (iii)
- D. (i) < (ii) < (iii)

42. The products of hydrolysis of



- A. $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO} + \text{CH}_3\text{CHO}$
- B. $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{CH}_3\text{CHO}$
- C. $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO} + \text{C}_2\text{H}_5\text{OH}$
- D. $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{C}_2\text{H}_5\text{OH}$

43. A compound A of formula $\text{C}_3\text{H}_6\text{Cl}_2$ on reaction with alkali can give B of formula $\text{C}_3\text{H}_6\text{O}$ or C of formula C_3H_4 . B on oxidation gave a compound of the formula $\text{C}_3\text{H}_7\text{O}_2$. C with dilute H_2SO_4 containing Hg^{2+} ion gave D of formula $\text{C}_3\text{H}_6\text{O}$, which with bromine and NaOH gave the sodium salt of $\text{C}_2\text{H}_4\text{O}_2$. Then A is:

- A. $\text{CH}_3\text{CH}_2\text{CHCl}_2$
- B. $\text{CH}_3\text{CCl}_2\text{CH}_3$
- C. $\text{CH}_2\text{ClCH}_2\text{CH}_2\text{Cl}$
- D. $\text{CH}_3\text{CHClCH}_2\text{Cl}$

44. Identify the product (A) in following reaction series, $\text{CH}_3\text{CN} \xrightarrow{\text{Na/C}_2\text{H}_5\text{OH}} \text{(X)} \xrightarrow{\text{HNO}_2} \text{(Y)} \xrightarrow{[\text{O}]} \text{(Z)} \xrightarrow{\text{Tollens reagent}} \text{(A)}$:

- A. CH_3CHO
- B. CH_3CONH_2
- C. CH_3COOH
- D. $\text{CH}_3\text{-CH}_2\text{-NHOH}$

45. During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?

- A. Concentrated sulphuric acid
- B. Dilute nitric acid
- C. Dilute sulphuric acid
- D. Dilute hydrochloric acid

ANSWER KEY

Qn	Ans	Qn	Ans	Qn	Ans	Qn	Ans	Qn	Ans
1	D	10	B	19	B	28	D	37	B
2	B	11	B	20	C	29	C	38	A
3	D	12	A	21	A	30	B	39	A
4	A	13	D	22	A	31	A	40	C
5	D	14	A	23	C	32	C	41	B
6	B	15	B	24	B	33	B	42	C
7	C	16	D	25	C	34	C	43	A
8	A	17	A	26	A	35	A	44	C
9	C	18	B	27	A	36	A	45	C